I have cloned this GitHub repository for my individual Vibe Coding assignment:
SETAPESU26/04_target_shooting.git
I have already cloned it to my local machine.
Please analyze the project structure and README requirements. I need to fix the existing bug and implement every feature requested in the README.
For now, DO NOT modify or rewrite the code. First:

1. Identify the existing bug.

Show more

I checked the repository and its README. The repo currently has one commit and contains the `target-shooting` project folder plus the README and `.gitignore`.

1. Existing bug

The deliberate bug is in **hit detection**.

The README says `Target.get_bounding_rect()` in:

```
target-shooting/game/target.py
```

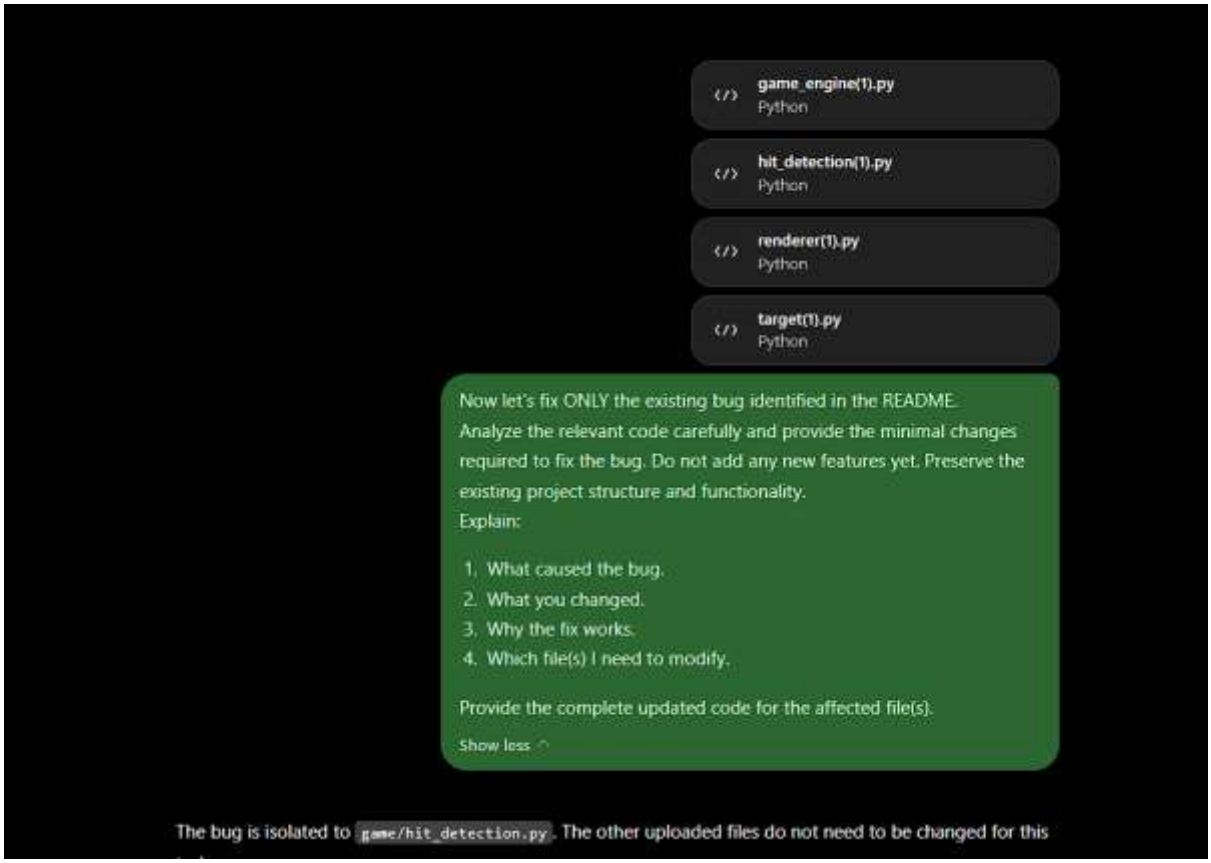
currently creates the rectangle approximately as:

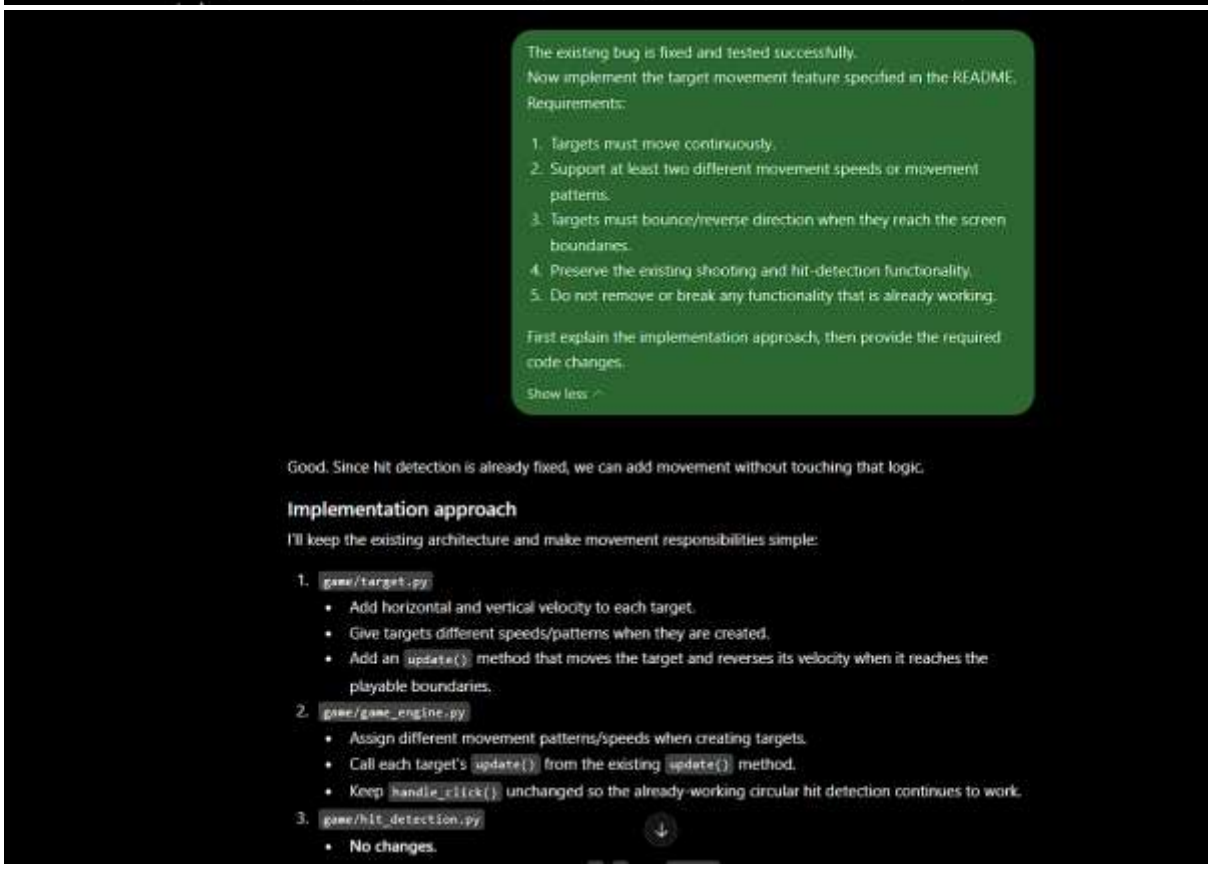
```
pygame.Rect(self.x, self.y, radius*2, radius*2)
```

But `self.x, self.y` represent the **center** of the circular target, whereas `pygame.Rect` expects its first two values to represent the **top-left corner**.

Therefore, the hitbox is shifted down and right relative to the visible circle.

More importantly, the assignment explicitly asks you to fix the actual hit test in:


game_engine(1).py
Python
hit_detection(1).py
Python
renderer(1).py
Python
target(1).py
Python
Now let's fix ONLY the existing bug identified in the README.
Analyze the relevant code carefully and provide the minimal changes required to fix the bug. Do not add any new features yet. Preserve the existing project structure and functionality.
Explain:
1. What caused the bug.
2. What you changed.
3. Why the fix works.
4. Which file(s) I need to modify.
Provide the complete updated code for the affected file(s).
Show less
The bug is isolated to game/hit_detection.py. The other uploaded files do not need to be changed for this


The existeng bug is fixed and tested successfully.
Now implement the target movement feature specified in the README.
Requirements:
1. Targets must move continuously.
2. Support at least two different movement speeds or movement patterns.
3. Targets must bounce/reverse direction when they reach the screen boundaries.
4. Preserve the existing shooting and hit-detection functionality.
5. Do not remove or break any functionality that is already working.
First explain the implementation approach, then provide the required code changes.
Show less
Good. Since hit detection is already fixed, we can add movement without touching that logic.
Implementation approach
I'll keep the existing architecture and make movement responsibilities simple:
1. game/target.py
• Add horizontal and vertical velocity to each target.
• Give targets different speeds/patterns when they are created.
• Add an update() method that moves the target and reverses its velocity when it reaches the playable boundaries.
2. game/game_engine.py
• Assign different movement patterns/speeds when creating targets.
• Call each target's update() from the existing update() method.
• Keep handle_click() unchanged so the already-working circular hit detection continues to work.
3. game/hit_detection.py
• No changes.

The target movement feature is implemented and tested.

Now implement the score and combo multiplier feature specified in the README.

Requirements:

- Award points when a target is successfully hit.
- Implement the required combo multiplier.
- Display the score and combo information clearly.
- Ensure the scoring system integrates with the existing shooting and hit-detection logic.
- Preserve all previously implemented functionality.

Explain the changes and provide the required code.

Show less

According to the README, Task 3 requires that **each hit adds points based on the current combo multiplier, consecutive hits increase the multiplier, a miss resets the combo, and both score and multiplier are displayed during play**.

Since movement and circular hit detection are already working, we only need to extend the existing game-state and rendering logic. I would **not change** `target.py` or `hit_detection.py`.

Implementation approach

We'll make these changes:

- Add `score` and `combo` to `GameEngine`.
- Start the combo at `1x`.
- On a successful hit:
  - Award `10 * current_combo` points.
  - Increase the combo for the next hit.

The scoring and combo system is implemented and tested.

Now implement the 30-second countdown, end-of-round behavior, and restart functionality specified in the README.

Requirements:

- Start a 30-second countdown when the round begins.
- Display the remaining time.
- End the round when the timer reaches zero.
- Prevent normal gameplay after the round ends.
- Display the final result appropriately.
- Allow the player to restart the round.
- Preserve all previously implemented features.

Explain the changes and provide the required code.

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The cleanest way to implement Task 4 is to keep the **round state inside** `GameEngine` and let the existing game loop continue running for rendering, while disabling movement and clicks once the timer expires.

There is one important integration point: **restart input belongs in** `main.py`. I don't have your current `main.py`, so I won't invent its event-loop structure and risk overwriting your existing functionality. I can give you the complete `game_engine.py` change now, plus the small `main.py` change you need to insert into your existing event handling.

1. Implementation approach

The round will have three states conceptually: